



Paper-thin degradable batteries

Scientists from Singapore have manufactured paper-thin biodegradable zinc batteries.
<https://digit.in/jan22-43>



DNA molecules in ancient dirt

DNA molecules in ancient dirt offer a treasure trove of clues to our past.
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Blood consists of plasma, red blood cells (RBC), white blood cells (WBC) and platelets. Although all blood consists of the same basic elements, not all blood is the same. The WBC provides immunity and acts as resistance to viruses and infections. The “type” alludes to the presence of a specific type of antigen poking up from the surface of a red blood cell. An antigen is something that gets a reaction from an immune cell known as an antibody which works as an immunizer when exposed to bacteria or viruses. These antibodies attach themselves to the harmful microorganisms, cluster them together and evacuate from the system.

Your blood group is determined by the type of antigen present in the cell. For instance, if only antigen of type A is present in your blood group, then it is classified as blood type A, and vice versa for B. If both A and B are present, then it is of the type AB, whereas if there is an absence of both the antigen then it is O type blood. So, if the blood of a type A person were donated to a person with type B blood, then the receiver's immune system would perceive the unfamiliar sugars as an intruder and attack it, resulting in the death of the receiver.

The positive and the negative sign of your blood group indicates the presence or absence of an antigen known as the Rh factor. The Rh factor is an inherited protein found on red blood cells. Rh factor is significant in blood donations and transfusion. A person with a +Rh factor does not make -Rh antibodies but individuals with -Rh factor do make them. Therefore someone with +Rh blood group can receive blood from both Rh factors but those with -Rh factor can only receive blood from -Rh. If blood containing different antigens is transferred to a person, their immune system will fail to recognise the antigen and attack it.

This is why O negative blood type is termed as ‘universal donor’ since it does not have the molecules that would provoke a fatal reaction.

SOME FACTS ABOUT BLOOD GROUP

1. Every year blood transfusion saves more than 4.5 million lives.
2. Less than 1% of the population have AB- blood type. It's the rarest rare blood type.
3. Blood type is also inherited or passed genetically from our parents.
4. Blood group C was later changed to group O to indicate zero presence of antigens.



The discovery of different blood types raised many questions as to how and why different blood types or antigens came into being and do animals also possess different types of blood or is it a distinct human feature? It was in the 1900s when scientists studied the molecular biology of the blood they found in a single gene called ABO, which is responsible for the antigen in our blood type. People with blood type O have mutated ABO genes which hinder them from making the enzymes.

Our blood types might be older than we thought, however, it's difficult to tell how old. Researchers still can't seem to break down the qualities so they can't understand how and what created these differences. In any case, the proof that scientists have assembled up until this point uncovers a violent history to blood classifications. Chimpanzees, our nearest living family members, have just type A and type O blood. Gorillas, have just B. Now and again, mutations have modified the ABO gene, transforming type A blood into type B. And surprisingly in people, researchers are finding genetic mutations that have emerged over and over that keep the ABO protein from forming the antigen, and these have diverted blood classifications from A or B to O.

Studies have proved that blood types evolved around 20 million years ago, however, the cause and consequence are still studied. The

evolution of blood types can be attributed to illness or diseases. There are many coalitions found between our blood groups and infections, cancer and many life-threatening diseases. Our blood has over the years adapted itself to fight diseases and has built an immune system against it. According to Christine Cserti-Gazdewich, a haematologist at Toronto General Hospital, Malaria was the reason behind the evolution of the type O blood group. Type O is more dominant in Africa and countries with high cases of Malaria which indicates that type O blood contains some advantage against the disease.

Malaria cells infect cells with type A sugar to form clumps known as “rosettes”, which can be dangerous when they structure in vital organs including the brain. Therefore, people with blood type O are less susceptible to contracting malaria as they lack the sugar molecules and are less sick if they do get infected. Then again, people with type O might be more susceptible to different diseases. They are known to be more prone to *Helicobacter pylori*, a bacteria that causes ulcers and ruptured achilles tendons. Blood type A has a higher risk of contracting several types of cancer in the stomach, pancreas and leukaemia.

Your blood type has secrets that haven't been unearthed yet and there is also a possibility it has nothing to do with your blood and has a set purpose of its own. Only bloody time will tell. Pun intended. 